Example 4.6 In ΔLMN, LM is extended to O. If $\angle L = 62°$ and $\angle N = 31°$, find $\angle NMO$.

Solution

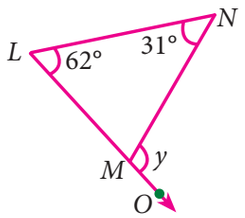


Fig. 4.13

Let, $\angle NMO = y$

Exterior angle = sum of two interior opposite angles

$$y = 62° + 31°$$

$$= 93°$$

Example 4.7 In the $\triangle ABC$ shown in the figure, find the angel z.

Solution

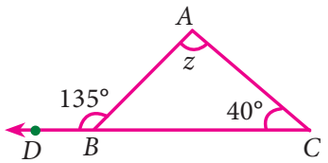


Fig. 4.14

Exterior angle = sum of two interior opposite angles

$$135° = z + 40°$$

Subtract 40° on both sides

$$135° - 40° = z + 40° - 40°$$

$$z = 95°$$

Example 4.8 In the given isoceles triangle IJK (Fig. 4.15), if $\angle IKL = 128°$, find the value of x.

Solution

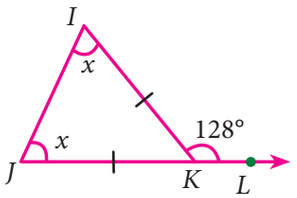


Fig. 4.15

Exterior angle = sum of two interior opposite angles

$$128° = x + x$$

$$128 = 2x$$

$$\frac{128}{2} = \frac{2x}{2} \quad \text{[on both sides, divide by 2]}$$

$$x = 64°$$

Example 4.9 With the given data in the Fig. 4.16, find $\angle UWY$.

What do you infer about $\angle XWV$?

Solution

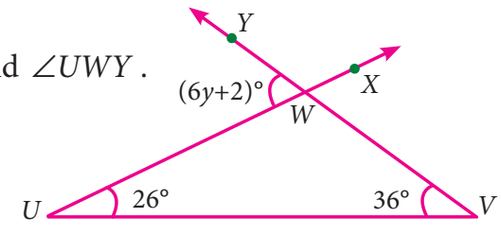


Fig. 4.16

Exterior angle = sum of two interior opposite angles

$$6y + 2 = 26° + 36°$$

$$6y + 2 = 62°$$

Subtract 2 from both sides

$$6y = 62 - 2$$

$$6y = 60°$$

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